

PRISMA 2020 Main Checklist

Topic	No .	Item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title page: "... A PRISMA 2020 Systematic Review of the Empirical Literature, 2015-2026".
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist	Abstract (structured; see PRISMA 2020 for Abstracts checklist below).
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Section 1 (Introduction) and Section 2 (Literature Background and Related Reviews; Table 1).
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Section 1, "Objectives": research questions RQ1-RQ4.
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Section 3.4 and Table 3 (inclusion/exclusion criteria); grouping into themes F, D and S for synthesis (Section 3.4; Appendix B).
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Section 3.2: Crossref REST API, last searched 5 October 2026; targeted Crossref title searches; Scopus/WoS not accessible (Section 6.1).
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Section 3.3, Table 2 and Appendix C (all 70 query strings, filters and limits).

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Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Section 3.5: two automated keyword stages (search/screen.py, screen_stage2.py), title/abstract assessment and full-text check of retrieved reports (21 of 42) by one reviewer (an AI agent); no independent duplicate screening.
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Section 3.6: one reviewer (AI agent) extracted data from abstracts and Crossref metadata; no confirmation from study authors.
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Section 3.6 and Table 4: outcomes coded per study (outcome, method family, benchmark, accuracy measure, forward projection, sustainability dimension); all abstract-reported measures collected.
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Section 3.6 and Table 4 (setting, data period, open-access status); items not reported in the abstract coded "n/r" and never inferred (Appendix B).
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Section 3.7: six-item reporting-transparency screen (abstract level), one reviewer; formal risk-of-bias tool (e.g. PROBAST) not applied - stated as a limitation (Section 6.1).
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Not applicable: no effect measures pooled; accuracy measures are described (Section 3.8; Figure 7).
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item 5)).	Section 3.8: studies grouped by theme (F, D, S) per eligibility coding (Section 3.4).

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	13 b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Section 3.6: missing items coded "n/r"; no data conversions.
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Section 3.8: descriptive figures (Figures 2-7), keyword network (Figure 5), study tables (Tables 5-7).
	13 d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Section 3.8: descriptive, keyword co-occurrence (controlled vocabulary, modularity clustering) and thematic synthesis; no meta-analysis (heterogeneous outcomes).
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Not performed (no meta-analysis); heterogeneity described narratively (Sections 4.3-4.4).
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Not performed; recall check of the automated screening reported (Section 6.1).
	Reporting bias assessment	14 Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Not performed (stated in Section 3.8).
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Not performed (stated in Section 3.8).
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Section 4.1 and Figure 1 (PRISMA flow diagram; supplementary completed template PRISMA_2020_flow_diagram_completed.docx).
	16 b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Appendix E lists all records excluded at title/abstract and full-text assessment with reasons.
Study characteristics	17	Cite each included study and present its characteristics.	Tables 5 and 6 (each included study cited with its characteristics); Figures 2-4.

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Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Section 4.6 (reporting quality of forecasting studies, abstract level); formal risk of bias not assessed.
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Tables 5-6 (abstract-level characteristics); effect estimates not extracted because full texts were not assessed (Section 6.1).
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Sections 4.3-4.4 and Table 7 (characteristics of contributing studies per theme).
	20 b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Sections 4.2-4.5: descriptive counts and cross-classification (Table 7); no statistical synthesis.
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Not applicable (no statistical synthesis).
	20 d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Not applicable (no sensitivity analysis).
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Not assessed (Section 3.8).
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Not assessed (Section 3.8).
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Sections 5.1-5.2 (principal findings; comparison with prior reviews).
	23 b	Discuss any limitations of the evidence included in the review.	Sections 5.1 and 6.1 (abstract-level evidence, missing abstracts, heterogeneous reporting).

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	23c	Discuss any limitations of the review processes used.	Section 6.1 (single database, capped retrieval, automated screening recall, single AI reviewer, partial full-text assessment).
	23 d	Discuss implications of the results for practice, policy, and future research.	Sections 5.3-5.5 (framework, practical implications, future research agenda; Table 8).
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Not registered (Section 3.1).
	24 b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	No protocol was prepared in advance; criteria and code archived in the repository (Section 3.1).
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Not applicable (no registration or protocol).
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Declarations: funding none declared [to be completed by the authors].
Competing interests	26	Declare any competing interests of review authors.	Declarations: none declared [to be completed by the authors].
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Declarations and Appendix D: search logs, screening outputs, extraction data (data/included.json) and all code in the repository directory review/.

PRISMA Abstract Checklist

Topic	No.	Item	Reported?
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Yes
Synthesis of results	6	Specify the methods used to present and synthesize results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Yes
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	Yes
Registration	12	Provide the register name and registration number.	Yes

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